

Models for Relocation of Emergency Medical Stations

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Abstract The paper presents a mathematical programming model for the deployment of stations where the ambulances are kept. The model deals with the reliability of service. Reliability in emergency systems is based on the individual level of service at each demand point. The system operates, if there is an available ambulance that can respond immediately to each emergency call. However, the probability that vehicles are available can hardly be determined before the stations are located. That is why an upper bound on the failure probability is used in presented location model instead of the probability itself. This upper bound is calculated using a Poisson distribution of arriving calls and the maximum time duration of a vehicle service trip. The model maximizes an average reliability that can be achieved on a given territory with a limited number of vehicles. Such a model may be useful when the issue is a reorganization of the current emergency system. A computer simulation model was used to verify vehicle location provided by the mathematical model. Both models were implemented under the conditions of the Slovak Republic. Travel times were calculated using the transportation network data from the OpenStreetMap project. The paper compares the proposed vehicle location with the current one in terms of the objective function and performance characteristics resulting from computer simulation.

Keywords Emergency medical service • Service reliability • Vehicle location • Integer programming • Computer simulation

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1 Introduction

The quality and efficiency of the emergency medical service (EMS) depends mainly on:

- the number of ambulances operating in a given region and
- the deployment of stations where the ambulances are kept.

Specifying the proper number of ambulances is a sensitive issue balancing between two opposing requirements. On one hand, the main task of the EMS—to save lives and reduce human suffering caused by injuries or illnesses—can be fulfilled if the network of emergency stations is sufficiently dense so that each emergency call could receive an immediate response and first aid could be provided in a very short time. On the other hand, there is a justified requirement on the efficiency of public expenditures that reduces the number of ambulances. Reducing the number of ambulances increases their workload and results in a lower availability of urgent health care since the nearest ambulance may be busy at the moment when a new call arrives. The situation when the patient does not receive medical care in a predefined time limit is interpreted as the system failure (Ball and Lin 1993).

This paper concerns the strategic level of the urgent health care management and aims at determining the optimal location and sizing of emergency stations so that the quality of service is as good as possible. Clearly, the main quality output of the EMS production is response time. However, due to the stochastic character of the system, it cannot be modelled directly and various approximation criteria are used instead. We measure the efficiency of the EMS by its reliability, which is defined as the probability that the service will be provided when and where necessary.

The problem of station location and determining the number of vehicles to allocate to each station is denoted as the EMS system design problem (Noyan 2010). The research presented here deals with the existing EMS system and looks for a better station location preserving the current number of vehicles. Therefore the problem to be solved can be expressed by the following question: Given the number of vehicles, where they should be located so that the average reliability of service over all demand nodes could be as high as possible?

We formulate the problem as an optimisation problem and solve it using a mathematical programming approach. The intention is to devise such a model that would be applicable in a wide-spread region containing both urban and rural areas. We have in mind a state territory. The motivation stems from the situation in the Slovak Republic where ambulances are kept in single-purpose stations. Every station houses only one ambulance. The number and locations of stations on the whole state territory are defined by the Regulations of the Ministry of Health of the Slovak Republic No. 10552/2009-OL and 11378/2010-OL. In accordance with the Regulations, 273 stations are currently deployed.

Because stations are supposed to be deployed in a large-scale area, a macroscopic view must be applied. It means that demands are aggregated and customers

of the system are not individual patients but whole municipalities. In accordance with Felder and Brinkman (2002), we suppose that the number of calls arising in a municipality is proportional to the number of its inhabitants and so we identify the size of the demand with the municipality population. At the same time, every municipality is regarded as a candidate location for a station. To simplify the model, we allow to place at most one station in a municipality and to allocate multiple vehicles to a station. Such a situation could be resolved in practice so that multiple stations equipped with one ambulance would be dispersed over the municipal area.

2 Literature Review

Ambulance location models are surveyed in Brotcorne et al. (2003). In this review paper, the models are classified into two main categories: deterministic and probabilistic models.

Deterministic models of the EMS systems are based on an implicit assumption that there is always an ambulance available to respond to a call. They can be further divided into covering and allocation models. In covering models, a maximum value is preset for either distance or travel time. If a service is provided by a facility located within this limit, then the service is considered acceptable, and a customer is considered covered by the service, if he or she has a facility sited within the preset distance or time. In the allocation models, demand zones are assigned to ambulance locations. One of the possible models of this type is the p -median model that minimizes the average travel time from the ambulance locations to potential patients. The objective function is a surrogate of the efficiency criterion which aims at providing the best possible level of service to as many people as possible with a limited number of resources. Besides efficiency, equity (or fairness) is a core performance dimension in a health care system (see e.g. Kvet and Janáček 2014; Janáček and Kvet 2015). Fairness is achieved when each customer receives the service of required and/or acceptable quality. Although an equity principle is usually favoured in health care provision, study by Felder and Brinkmann (2002) proves that efficiency maximizes the number of lives saved and is often applied by policy makers.

Probabilistic models take into account the reliability of service provision that may be affected by uncertainty present in a real system. Arriving calls are stochastic events, and treating a patient is also a random variable. Moreover, the travel time of an ambulance may be affected by the traffic and weather conditions. Therefore, the nearest ambulance may happen to be busy when an accident occurs. Then another ambulance must be dispatched to serve the call, or the service must be postponed. Probabilistic models incorporate so called busy fraction of ambulances, which is the probability that the ambulance will be busy when a new call arrives and it will not be able to respond to the call. For a thorough discussion of the probabilistic models see, for example, a chapter by ReVelle and Williams (2004). Sorensen and Church (2010) distinguish three generations of probabilistic models. First-generation

probabilistic models suppose that all ambulances are equally busy. Models are variants of the famous Maximum Expected Coverage Problem (MEXCLP) formulation proposed by Daskin (1983). The Maximum Availability Location Problem (MALP), introduced by ReVelle and Hogan (1989) represents the second generation of probabilistic models. These models abandon the assumption of the uniform distribution of workload among ambulances and take into account the workload of individual ambulances. The probability of an ambulance being busy is calculated as the ratio of the total demand (in hours) within the local service area surrounding the server and the available service time of all servers located within the same area. The number of calls that can be responded to with a predefined level of reliability is maximized. Models of the third generation [represented by the Queueing-based Maximum Availability Location Problem (Marianov and ReVelle 1996)] maintain the concept of individual busy-fractions but estimate them using the queueing theory.

Common drawbacks of these models are simplifying assumptions that are necessary to approximate probabilistic behaviour within linear equations. Some of the more significant assumptions include (Sorensen and Church 2010):

1. Server independence meaning that the probability of one ambulance being available to handle a call is independent on the status of other ambulances in the region.
2. Locally-constrained service areas meaning that ambulances within a local area only serve calls within the same area and vice versa demand nodes located within the local area are not served by ambulances from outside the area.
3. Location-independent service times meaning that service times do not depend on the distances between stations and call locations. This assumption prevents the application of the mentioned models in a large-scale network, where distances between stations and incident sites differ significantly.
4. Uniform service times that do not distinguish categories of patients, severity of their diagnosis nor a possible need of their transport to a hospital. Obviously individual demands cannot be distinguished in the system design, however if the models are based on an average service time, they need not give realistic outputs.

Exact service times are not included in the reliability model introduced by Ball and Lin (1993), just the upper limit of the service time inputs the model and is used to calculation of the upper bound on the failure probability. This upper bound is based on the assumption that nobody outside the service area can help to serve demands within the area. So the upper bound remains in force also in the case of overlapping areas where the servers can help each other. If locally-constrained service areas do not represent an obstacle to the applicability of the reliability upper bound, neither does the server independence assumption. The consequence of the first and second simplifications is that the model takes into account a more pessimistic failure probability. However, the model does not need other parameters estimation of which would be difficult, and that is why we took it as a basis for our

vehicle relocation model. Although Ball and Lin's model is often classified only as a variant of the MALP, we think that they propose a principally new approach. They do not calculate the probability of an ambulance's unavailability as a ratio of demanded and offered service times but as the probability that the number of calls arising in the service areas of the servers located in the coverage area of the given demand point exceeds the number of available ambulances. Ball and Lin solve the Reliability Location Set Covering Problem that minimizes the number of vehicles needed to ensure a desired level of reliability for each demand node. The model was further elaborated by Borrás and Pastor (2002) who regarded the EMS system as a system with losses of calls in case of congestion. The reliability constraints proposed by Ball and Lin were also used by Beraldi and Bruni (2009) to improve the two-stage EMS model that incorporates first-stage strategic decisions about location and capacities of the EMS stations and second-stage tactical decisions concerning assignment of demand nodes to stations.

3 Mathematical Programming Model

In this section we describe a mathematical programming model that aims at finding a new location of the current number of ambulances. The model is based on the definition of the failure probability introduced by Ball and Lin (1993).

It is supposed that potential station locations have been predetermined due to some previous studies. Let us denote the set of location candidates by symbol I . Its elements are indexed by symbol i . Potential patients live in municipalities spatially spread in the given territory. We denote the set of municipalities (demand points) by J and the population of municipality j by b_j . Members of both sets I and J correspond to nodes of a road network. The network segments are associated with the travel times that depend on the length of the segment, category of the road and average speed of an ambulance. Based on the segment travel times, the shortest travel time t_{ij} from node $i \in I$ to node $j \in J$ can be calculated using, for example, the Dijkstra's algorithm. The travel times are needed to develop a set of candidate locations that cover each municipality j , it means, that are within the specified response time T^{max} of municipality j . We denote this set by N_j and define $N_j = \{i \in I: t_{ij} \leq T^{max}\}$. This specification of the problem shows that we face a network location problem (Daskin 2013).

The acceptable level of service is ensured by imposing an upper limit on the system failure for each demand point. The system fails if there are no feasible vehicles available for the service when a demand call occurs. The failure probability should be less than $1 - \alpha$, where α is a minimum level of reliability. Due to the difficulty of computing the exact failure probability, Ball and Lin developed an upper bound on it. A patient cannot be served immediately if there is no available

vehicle in covering stations at time t when the emergency call arrives. This situation occurs if all vehicles have been servicing other demand calls. To calculate the probability that all feasible vehicles are busy at time t we need to know the duration of service trips. Since an upper bound on the failure probability is the issue, a time amount T equal to an upper bound on service time may be used. The probability that the number of calls that have occurred in the previous interval of size T in the service area of station i exceeds the number of available vehicles can be denoted as $\Pr(D_i \geq K_i)$, where D_i is the number of calls that have occurred in the service area of station i during time interval $\langle t - T, t \rangle$ and K_i is the number of vehicles housed at station i . If calls arise from each demand point according to the Poisson distribution, then the upper limit of the system failure for demand point j is defined by the term on the left hand side of the following relation:

$$\prod_{i \in N_j} \Pr(D_i \geq K_i) \leq 1 - \alpha \quad (1)$$

Although the product form suggests that service areas of the stations do not overlap, it still remains an upper bound in a more realistic case of overlapping areas.

In the location model, the number of vehicles housed at each station is a decision variable that can be defined as follows.

$$x_{ik} = \begin{cases} 1 & \text{if } k \text{ vehicles are stationed at location } i, \\ 0 & \text{otherwise,} \end{cases}$$

for $i \in I$ and $k = 1, 2, \dots, M_i$, where M_i is the maximum number of vehicles that can be located at i . Using these variables the restriction (1) can be written as

$$\prod_{i \in N_j} \prod_{k=1}^{M_i} [\Pr(D_i \geq k)]^{x_{ik}} \leq 1 - \alpha \quad (2)$$

By taking the logarithm function on both sides and changing the signs, (2) can be transformed into a linear constraint:

$$\sum_{i \in N_j} \sum_{k=1}^{M_i} a_{ik} x_{ik} \geq -\log(1 - \alpha) \quad \text{for } j \in J \quad (3)$$

where $a_{ik} = -\log(\Pr(D_i \geq k))$.

The optimization criterion will be the weighted average of failure probabilities over all municipalities where the weight is the population of municipality j . Let α_j be the unknown reliability in municipality j . To simplify the notation we introduce a new variable y_j that substitutes for $-\log(1 - \alpha_j)$. It represents an upper bound on the failure probability in municipality j on the logarithmic scale.

The complete integer programming formulation of the model is as follows:

$$\text{maximize } \frac{1}{\sum_{j \in J} b_j} \sum_{j \in J} b_j y_j \quad (4)$$

$$\text{subject to } \sum_{k=1}^{M_i} x_{ik} \leq 1 \quad \text{for } i \in I \quad (5)$$

$$\sum_{i \in N_j} \sum_{k=1}^{M_i} a_{ik} x_{ik} \geq y_j \quad \text{for } j \in J \quad (6)$$

$$\sum_{i \in I} \sum_{k=1}^{M_i} k x_{ik} \leq p \quad (7)$$

$$y_j \geq -\log(1 - \alpha) \quad \text{for } j \in J \quad (8)$$

$$x_{ik} \in \{0, 1\} \quad \text{for } i \in I, k = 1, 2, \dots, M_i \quad (9)$$

Constraints (5) state that each location i either is not chosen or houses exactly one number of k vehicles. Constraints (6) bound new variables y with location variables x . Constraint (7) ensures that the number of allocated vehicles does not exceed a predefined number p . Constraints (8) are lower bounds on variables y ensuring a minimum level of reliability α at each demand node j . Constraints (9) are obligatory constraints on bivalent variables.

To calculate the average failure probability we must use the exponentiation to return the computation to the original scale. Finally instead of failure probability we can express the average lower bound on reliability $\bar{\alpha}$ per inhabitant using the following formula:

$$\bar{\alpha} = 1 - \exp \left(- \frac{1}{\sum_{j \in J} b_j} \sum_{j \in J} b_j y_j^* \right) \quad (10)$$

where y_j^* are optimal values of variables y in the solution of the model (4)–(9).

The average lower bound on reliability per municipality is:

$$\bar{\alpha} = 1 - \exp \left(- \frac{1}{n} \sum_{j \in J} y_j^* \right) \quad (11)$$

where n is the number of municipalities ($n = |J|$).

The fact that a station houses multiple vehicles can be interpreted in such a way that multiple stations equipped with one ambulance are dispersed over the town area. We will use this interpretation in the following sections describing a case study of the Slovak Republic. Moreover we will use a simplified term reliability instead of a lower bound on reliability.

4 Computer Simulation Model

For the verification of the mathematical model we created a microscopic simulation model. The simulation allows testing many variants of the location of stations. The agent based simulation model simulates all processes of emergency medical service. It was built in AnyLogic simulation software (Borschev 2013; Grigoryev 2014). AnyLogic is developed on Java simulation core. We implemented a library of classes and functions in Java for the simulation support. The main advantage of our simulation model is a very precise simulation of moving the ambulance. The OpenStreetMap (OSM) data (<https://www.openstreetmap.org>) are used for creating the transportation network (explained in Sect. 5).

During the simulation run, the shortest path in terms of travel time is calculated for every ambulance trip. The travel time through a road segment depends on the length of the segment and the speed of the ambulance. The OSM data distinguish six road categories and for every category we define a speed limit. The ambulance can move in two modes: when it travels to a patient or transports the patient to a hospital, then it drives at a higher speed as usually. We also take into account maximal speed limits from the OSM data file. The result of this solution is a very realistic model of the ambulance movement.

For modeling a patient position we apply a sophisticated strategy using data from the LandScan database (<http://web.ornl.gov/sci/landscan>). LandScan data represent an ambient population (average over 24 h) over the whole state territory. The territory of the Slovak Republic is covered by 70,324 grid elements.

A special population tree is built from the population grid. It helps to select one node of the transportation network as a source of the emergency call. The proposed population tree has six levels. From leaves (bottom) to the root (top) there are levels with OSM nodes, population grids, municipalities, districts, regions and the state (the root of the tree). A weight is associated with every node of the tree. The OSM nodes at the bottom level have zero weight. They are assigned to population grids according to their geographic position. The weight of a population grid is set to the number of people living in the grid (the value of the corresponding element in the LandScan database). On the next level we assign population grids to municipalities also according to a geographic position. The weight of a tree node representing a municipality is a sum of the weights of the child nodes. Using the same algorithm we build levels with districts and regions. When we generate a patient position, we

traverse from the root of the tree to the leaves. On every level we need to decide which child of the current node should be used. We generate a random number from the discrete uniform probability distribution (from zero to the weight of the current node) which is interpreted as a cumulative weight of some of its children. This way a child is chosen randomly but with regard to the population density. On the bottom level we take a random leaf. This strategy for location of patients in the space is very realistic.

For every simulation model the maximal simulation speed is very important. When the simulation run is slow, it is difficult to take enough data from it. Our model works with big data files containing population distribution, transportation network and borders (municipalities, regions). We maximize the speed of simulation by using our special algorithms and data collectors.

For example for searching the shortest path on the network we performed many experiments with variations of the Dijkstra's algorithm. We tested several data structures for storing the nodes and we measured their performance during the simulation run. The outcomes from the profiler for 50,000 calculations of the shortest path are presented in Table 1. In the first three rows there are computation times for the implementations where references to nodes are stored in HashMap, ArrayList, and LinkedList data structures, respectively. Here all nodes were processed during the initialization phase of the algorithm. However, the initialization of all nodes is time consuming. Therefore we proposed an efficient implementation, where BitSet is used for storing information about the node initialization. Every bit in BitSet represents a boolean value indicating, whether the corresponding node has already been initialized. The BitSet is cleared at the beginning. It is extremely fast operation. When the Dijkstra's algorithm encounters a node, it checks in the BitSet, whether the node is initialized. If not, the algorithm initializes it and sets the corresponding bit in the BitSet. This solution allows to initialize only visited nodes. In the last row of the table the computation time for the combination of the BitSet and HashMap is reported.

From the simulation we take a lot of statistical information:

- reliability for cities, regions, whole state,
- statistics about response time, ambulance workload, duration of the ambulance move (all for cities, regions, whole state),
- statistics about utilization of rescue stations,
- many statistics for hospitals and rescue stations.

The simulation model was validated using actual positions of rescue stations. On the basis of published data about emergency medical service in the Slovak Republic

Table 1 Total time spent for searching the shortest path

Data structure	Total time (ms)
HashMap	382,626
ArrayList	325,839
LinkedList	247,268
BitSet + HashMap	230,304

we can compare results from the model with the real data. Animation during the simulation allows the user to verify behaviour of the ambulance. The user can see also many statistics, states and positions of ambulances, patients etc. Our model is fully valid and it produces relevant data. The results from the simulation model are more relevant than from the optimization model, thanks modelling occupation of the ambulance.

5 OpenStreetMap

The OpenStreetMap is a service, which allows users all over the world to create their own maps. All maps are editable and users can use these geographical data for their own purposes (data are under Open Database License). The OpenStreetMap data are exported in xml (extensible markup language) format. The basic components of the OSM are nodes (defining points in space), ways (defining linear features and area boundaries), and relations (which are sometimes used to explain how other elements work together). We use nodes and ways for the simulation model. A node represents a specific point on the earth surface defined by its latitude and longitude. Each node comprises at least an id number and a pair of coordinates. A way is an ordered list of nodes that define a polyline. Ways are used to represent linear features such as roads.

The main reasons for the use of the OSM data are:

- open and free data with high precision,
- the AnyLogic software supports the OSM infrastructure,
- the database contains a lot of supplementary information (type of the road, maximal speed, one-way direction etc.).

From the OSM we can take also information about borders of municipalities, regions etc. Before the start of the simulation run, the program loads all data into internal data structures. The whole transportation network saved in the OSM data file includes 38,082 ways. To build the model of the transportation network, we transform the ways to the edges. The model is then a graph with 359,947 nodes and 364,862 edges.

Map data from OSM for the Slovak Republic contain some errors and we had to correct them. The main correction consisted in creating one continuous graph representing the transportation network. Some roads are not connected properly. If such a situation occurred, the simulation model could not find the route between two nodes. So we checked the data and created a continuous graph. The second problem is that some roads have a wrong one-way direction setting. These data cause an error, that the algorithm can't find the route to the village. If the shortest path is not found, the algorithm ignores one-way direction on this critical section.

6 Case Study

The mathematical programming model (4)–(9) was implemented by using the real data of the Slovak transportation network. The *Xpress* (2011) optimisation tool was used for optimisation.

Currently 273 stations are deployed in the area of the Slovak Republic. They are located in 211 towns and villages including urban districts of the capital Bratislava and the second largest Slovak town Košice. They altogether serve 2,926 municipalities populated with 5,410,827 inhabitants. Large towns have multiple stations. The most stations (5) are in Bratislava—Petržalka, an urban district of Bratislava, and in Prešov, a highly populated town in the eastern Slovakia.

The mathematical model preserves the total number of stations and looks for a better location of them in terms of reliability. The coefficients in the model were set in the following way:

Potential locations (candidates) for stations consist of all municipalities and all the nodes with existing EMS stations defined by the official regulations.

To create the set N_j of candidate locations that are in the neighbourhood of municipality j , a requested response time T^{max} must be defined. The Slovak legislation does not specify performance standards, there is only a recommendation to deliver medical care in 15 min from when a call for rescue arrives. If we neglect the time taken by a dispatcher to handle a call, T^{max} may be set to 15 min.

To calculate coefficients a_{ik} we need to know the maximum service time T and the arrival rate of calls in municipalities located in the service area of a candidate i . On the basis of the news and published analysis on the EMS performance in Slovakia, we consider $T = 100$ min as a reasonable estimation of the maximum time when an ambulance is busy. The number of calls that have occurred in the previous interval of size T in a municipality j is calculated using the Poisson distribution with arrival rate λ_j (calls per T minutes). The statistics describing the number of calls with particular municipalities were not available to us. Therefore, we suppose that the number of calls in a municipality is proportional to the number of its inhabitants. We stem from the aggregated statistics for the Slovak Republic mapping the year 2013, reporting that the overall number of patients was 465,076. We calculate the rate λ per one inhabitant, and use it to estimate $\lambda_j = \lambda b_j$, where b_j is the number of inhabitants registered in the municipality j .

The maximum number of vehicles that can be located at candidate i takes into account the population of the candidate and the size of the service area of one ambulance. According to the analysis of the EMS system (Bahelka 2008) the capacity limit of one ambulance is 25,000 inhabitants. It follows that $M_i = \lceil b_i/25000 \rceil$.

The remaining coefficients are $p = 273$ and the minimum reliability $\alpha = 0.3$.

A new location of 273 stations was calculated using the model (4)–(9). Currently there are 360 villages whose reliability is 0 because there is no station in their

15 min vicinity. In total 472 municipalities have reliability under the threshold value 0.3. The average reliability per municipality is 0.776, average reliability per inhabitant is 0.772. The model calculates the average reliability per municipality of 0.779 and the average reliability per inhabitant of 0.754. The model ensures that a minimum reliability of 0.3 will be maintained in every municipality. There are only 32 municipalities with this minimum reliability. The expected reliability of the particular municipalities is calculated by using optimal values of variables x in the left hand side of the inequality (3). Expected reliabilities are visualized on the map of the Slovak Republic in Fig. 1 for the current and proposed location of the EMS stations, respectively.

We used computer simulation to estimate performance characteristics of the proposed system more realistically. Out of the performance measures in the emergency systems we are most interested in those reported in Table 2. Reliability is calculated as the percentage of calls responded to within 15 min. Simulated reliability for the current and proposed location of the EMS stations is in Fig. 2.

The simulation reveals that the proposed mathematical programming model does not outperform the current design of the EMS system in the average response time, the number of municipalities with the average response time greater than 15 min,

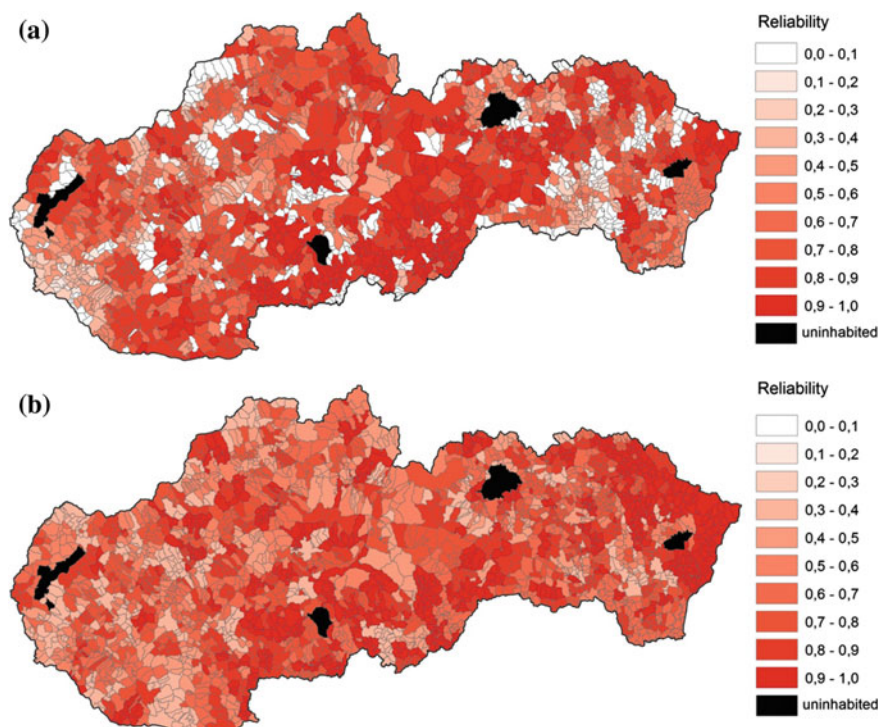


Fig. 1 Expected reliability: current location of the EMS stations (*above*), proposed location of the EMS stations (*below*)

Table 2 Performance characteristics from simulation

Output	Current location	Proposed location
Average response time	10.50	12.58
Number of municipalities with average response time >15 min	422	473
Average reliability	0.785	0.695
Number of municipalities with reliability <0.3	769	696

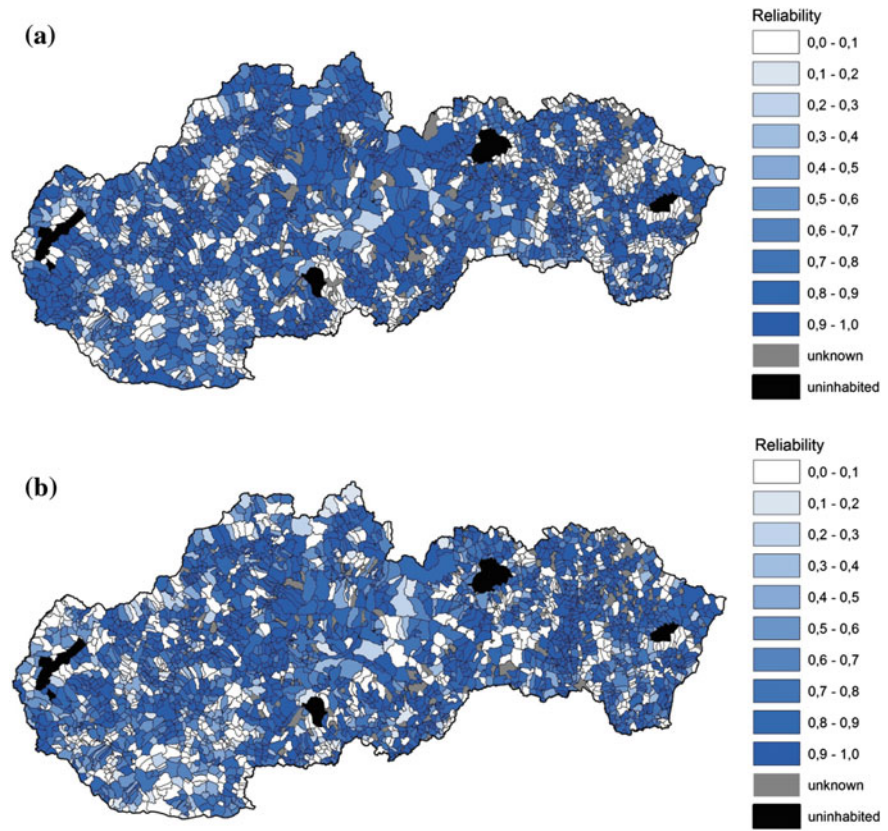


Fig. 2 Simulated reliability: current location of the EMS stations (*above*), Proposed location of the EMS stations (*below*)

neither in average reliability. However the model accentuates fairness as can be seen from the figures. It ensures that the reliability in more villages will be above the minimum level of 0.3.

7 Conclusions

In the paper, the problem of EMS ambulance location is formulated as a probabilistic problem that maximizes average reliability of the service over the state territory. We suggest that probabilistic models are more suitable than deterministic models in public service systems because they enable to model to some extent the uncertainty in the service provision. However in a real system there still exist many sources of the uncertainty that cannot be captured in a mathematical model. In such a situation a computer simulation model plays an undoubtedly significant role in the estimation of the performance characteristics of the system. The simulation model itself is not able to propose the best station location, however it is useful in the sensitivity analysis to answer the “what if” questions. It means it enables to experiment with changing system parameters such as the number of available ambulances in the proposed stations, demographic characteristics that influence the rate of calls in some regions and so on. Both models can serve as a decision supporting tool for an authority responsible for the emergency medical system design. A crucial issue that limits their usage is the input data relevant to the system operation. Reliable input data may improve the calibration of the models and their validation.

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